Consider the following game in matrices:

Two individuals, A and B, are engaged in a game of one nonsuitable game, game 1, or game 2. The payoffs are as in the game 1 or game 2 below.

Game 1 is played with probability x_1 and game 2 is played with probability x_2. The game is played between A and B, and the payoffs are as in the game 1 or game 2 below.

Game 1 is played with probability x_1 and game 2 is played with probability x_2. Let x_1 be the proportion of the population using game 1, x_2 be the proportion using game 2.

Game 1

	Player A	Player B
Strategy 1	1	1
Strategy 2	$1+2x_1$	$1+2(1-x_1)$

Game 2

	Player A	Player B
Strategy 1	$1+2x_1$	$1+2(1-x_1)$
Strategy 2	1	1

1. Write the replicator dynamics equation for the population with two types.
2. Write the equation in the form of a differential equation.
3. Define a given population.
4. Define an evolutionarily stable strategy (ESS).
5. Show that a given population is an ESS if and only if it is a strict equilibrium.
6. Consider the game given in the table below. Find the fixed points and determine which of them are ESS. Show your work.
7. Consider the game given in the table below. Find the fixed points and determine which of them are ESS. Show your work.

	Player A	Player B
Strategy 1	$1+ax_1$	$1+a(1-x_1)$
Strategy 2	$1+a(1-x_1)$	$1+ax_1$

Note: the parameter a is a constant, $0 < a < 1$.

Note: the parameter a is a constant, $-1 < a < 0$.

1. $f_A(x) = \underbrace{1}_{\text{base utility}} + \underbrace{2x_1}_{\text{gain utility from this game}}$

$f_B(x) = \underbrace{1}_{\text{base utility}} + \underbrace{2(1-x_1)}_{\text{gain utility from this game}}$

2. $\frac{dx_1}{dt} = x_1(f_B(x) - \phi)$ $\quad \frac{dx_1}{dt} = x_1(f_B(x) - \phi)$

where

$\phi = x_1 f_A(x) + (1-x_1) f_B(x)$

3. For a given population game a vector x^* is stable if

$\frac{df_i}{dx_i} = 0 \quad \forall \; i$

4. Given a population x_1 & given some $\varepsilon > 0$ and a strategy y the post entry population x_ε is:

$x_\varepsilon = x_1 + \varepsilon(y - x_1)$

5. A strategy x^* is an ESS if $\forall$ $x_1 \neq x^*$ sufficiently close to x^*.

$f(x^*, x^*) > f(x_\varepsilon, x^*)$

6. We have:

$f_A(x) = 1 + 2x_1 \qquad f_B(x) = 1 + 2(1-x_1)$

$\phi = x_1 f_A(x) + (1-x_1) f_B(x)$

$= x_1(1+2x_1) + (1-x_1)(1+2(1-x_1))$

$= x_1 + 2x_1^2 + (1-x_1)(3-2x_1)$

$= x_1 + 2x_1^2 + 3 - 5x_1 + 2x_1^2$

$= 3 - 4x_1 + 6x_1^2$

so

$\frac{dx_1}{dt} = x_1(f_A(x) - \phi)$

$= x_1(1+2x_1 - (3 - 4x_1 + 6x_1^2))$

$= x_1(-2 + 6x_1 - 6x_1^2)$

$= 2x_1(1-x_1)(2x_1 - 1)$

The fixed points are solutions of $\frac{dx_1}{dt} = 0$

$x_1 \in \{0, \frac{1}{2}, 1\}$

7. We use

$\frac{dx}{dt} = -2x_1(2x_1-1)(x_1-1)$

Euler's method:

$x_1^{(1)} = x_1^{(0)} + h \left.\frac{dx}{dt}\right|_{x^{(0)}}$

$x_1^{(0)} = 0.01$

$\left.\frac{dx}{dt}\right|_{x=0.01} = -2 \cdot 0.01(2 \cdot 0.01 - 1)(0.01 - 1)$

$\simeq -0.019404$

Thus

$x_1^{(1)} = 0.01 + 0.01 \cdot (-0.019404) \simeq 0.009806$

this decrease from a position close to $x_1 = 0$ indicates that $x_1 = 0$ is an ESS.

$x_1^{(0)} = .49$

$\left.\frac{dx}{dt}\right|_{x=0.49} = -2 \cdot 0.49(2 \cdot 0.49 - 1)(0.49 - 1)$

$\simeq -0.009996$

$x_1^{(1)} = 0.49 + 0.01(-0.009996) \simeq .489900$

This decrease from a point slightly below $x = \frac{1}{2}$ indicates that $x = \frac{1}{2}$ is NOT an ESS.

$x_1^{(0)} = 0.99$

$\left.\frac{dx_1}{dt}\right|_{x_1 = .99} = -2 \cdot .99(2 \cdot 0.99 - 1)(0.99 - 1)$

$\simeq 0.019404$

$x_1^{(1)} \simeq .99 + 0.01(0.019404) \simeq .9901 94$

This increase from a point just below $x_1 = 1$ indicates that $x_1 = 1$ could be an ESS.

7. $f_A(x) = 1 + ax_1 \qquad f_B(x) = 1 + a(1-x_1)$

$\phi(x) = x_1 f_A(x) + (1-x_1) f_B(x)$

$= x_1(1 + ax_1) + (1-x_1)(1 + a(1-x_1))$

$= 2ax_1^2 - 2ax_1 + a + 1$

$f_A(x) - \phi(x) = (1 + ax_1) - (2ax_1^2 - 2ax_1 + a + 1)$

$= a(-2x_1^2 + 3x_1 - 1)$

$= -a(x_1 - 1)(2x_1 - 1)$

$\frac{dx_1}{dt} = -x_1 a(x_1 - 1)(2x_1 - 1)$

The stable population is given by

$\frac{dx_1}{dt} = 0$

so $x \in \{0, \frac{1}{2}, 1\}$

For $a > 0$

$\frac{dx_1}{dt} = -ax_1(2x_1-1)(x_1-1)$

- for $0 < x_1 < \frac{1}{2}$:

$x_1 > 0, \; 2x_1 - 1 < 0 \; \; x_1 - 1 < 0$ so

$-ax_1(2x_1-1)(x_1-1) < 0$

- for $\frac{1}{2} < x_1 < 1$:

$x_1 > 0, \; 2x_1 - 1 > 0 \; \; x_1 - 1 < 0$ so

$-ax_1(2x_1-1)(x_1-1) > 0$

So for $a > 0$

$x_1 = \frac{1}{2}$ is unstable

$x_1 \in \{0, 1\}$ are stable.

For $-1 < a < 0$: we have a single multiplicative change of sign so

$0 < x_1 < \frac{1}{2} \quad \frac{dx_1}{dt} > 0$

$\frac{1}{2} < x_1 < 1 \quad \frac{dx_1}{dt} < 0$

$x_1 = \frac{1}{2}$ is stable

$x \in \{0, 1\}$ are unstable.